

I built a lightweight code search tool that lets people explore a codebase using natural language instead of exact keyword search. The idea is pretty simple: instead of needing to know the exact function or file name, you can just ask a question like you normally would, and it finds the most relevant parts of the code.

I wanted to build this because I've personally struggled with getting up to speed in new or large codebases. A lot of the time, the hardest part isn't writing code, it's just figuring out where things are and what everything is called. Traditional search only works if you already know the right keywords, which isn't always the case, so I wanted something that feels more intuitive.

On the backend, the tool reads through files in a project and turns them into embeddings using a transformer model. I store those in a FAISS index so searches stay fast. When a user types a query, it also gets turned into an embedding and matched against the code to find the closest results. I also added a small context feature so that previous queries can slightly influence the next one, which makes the experience feel more connected.

I focused a lot on making it feel fast and usable. It returns quick results first, then loads more detailed context after, so you're not just waiting around. I also added an overview mode that gives a higher-level summary of the codebase, which is helpful when you're just starting out and don't know what to look for yet.

I think this connects really well with what Pacific is working on. From what I understand, a big part of your work is helping people interact with complex information in a more natural way, and that's exactly the problem I was trying to solve here. This project got me really interested in semantic search and context-aware systems, and I'd love to keep building in that direction.

Overall, this was a really fun project and I learned a lot, especially about how important speed and context are when building tools people actually want to use.